INTRODUCING OUR SYSTEM

CLARA

AI-Powered Accounting Office Automation System

Graduation Project

Supervisor: Dr. Mohamed El Halaby

Made with GAMMA

THE CHALLENGE

Problem Statement

Accounting and administrative offices today struggle to keep operations organized as they grow:

1. **Task & File Management**

   Work and files are scattered with no structured way to assign or track them.

2. **Tracking Employee Work**

   Managers lack visibility into who is working on what, across branches and teams.

3. **Frequent Contract Updates**

   Updating contracts is slow and manual, consuming hours of staff time.

Result: poor communication, wasted time, and management overhead that slows the whole office down.

PROBLEM 01

Task Management Problem

!

Bad Communication

Employees and managers rely on WhatsApp chats, groups, or direct calls — making communication and tracking work much harder.

!

No Dedicated Workspace

All task discussion spreads across chat history, making it hard to see who is doing what and what's actually done.

Outcome: tasks fall through the cracks, deadlines slip, and managers can't intervene in time.

SOLUTION

A Dedicated Task Management System

CLARA introduces a specialized task management workspace between managers and employees to track work clearly and consistently.

Employee View

Add comments and attach files to communicate progress directly on the task.

Manager View

See assigned tasks and status, expand to review comments/files, and accept or reject work with feedback.

Notifications

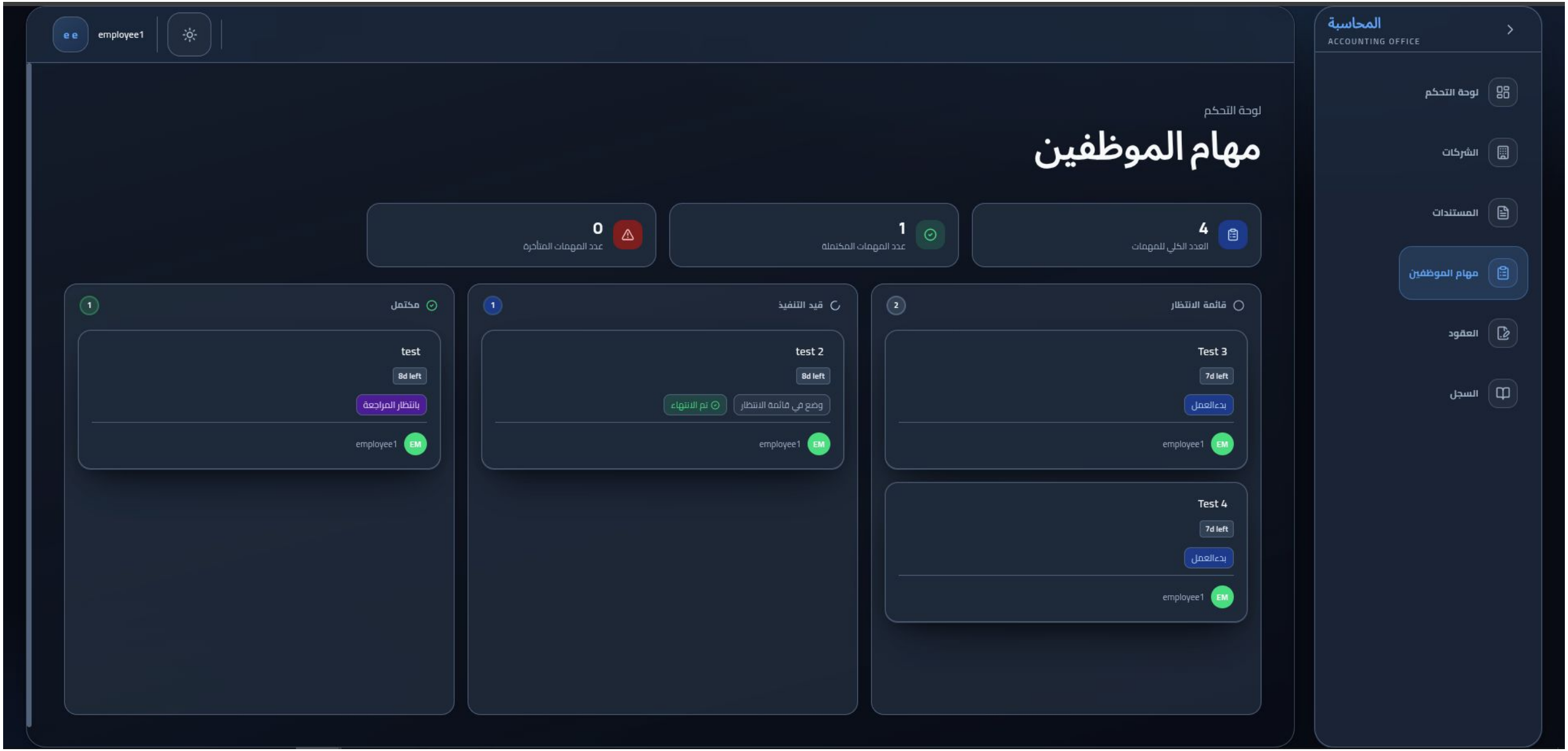
Both sides are notified instantly when a task they're engaged in is updated.

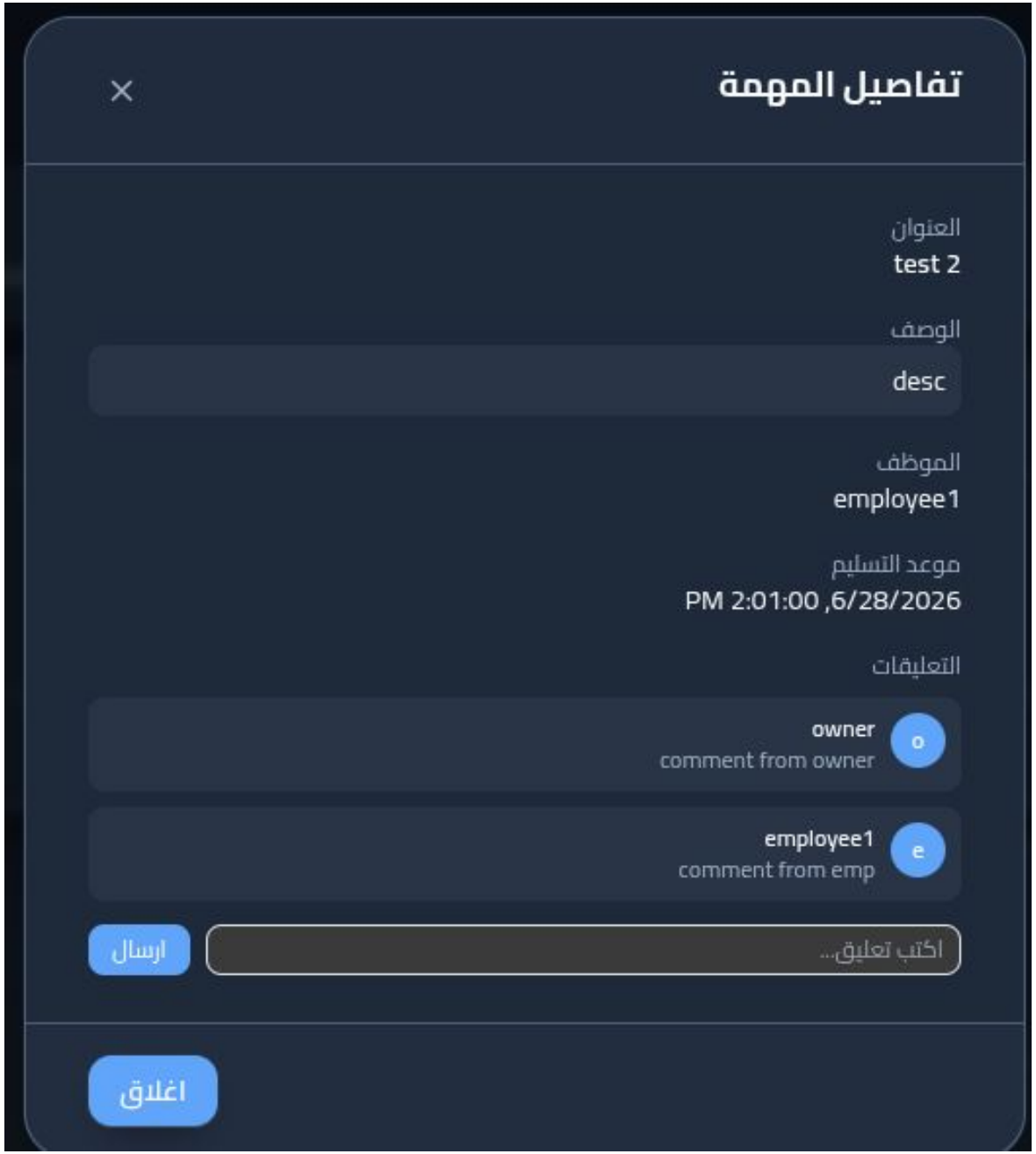
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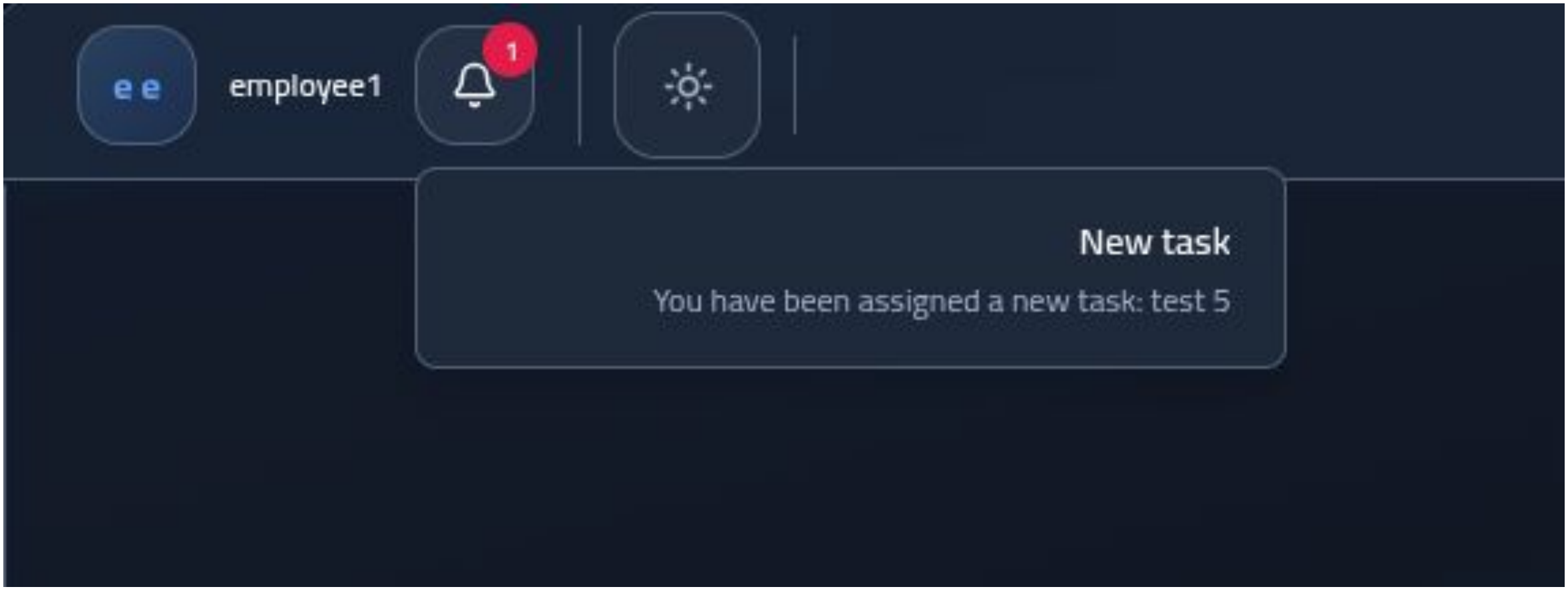
These features directly solve the task-tracking problem introduced earlier.


ee
employee1
المحاسبة
ACCOUNTING OFFICE
لوحة التحكم
الشركات
المستندات
مهام الموظفين
العقود
السجل
لوحة التحكم
مهام الموظفين
4
العدد الكلي للمهمات
1
عدد المهمات المكتملة
0
عدد المهمات المتأخرة
قائمة الانتظار
2
Test 3
7d left
بدء العمل
EM
employee1
Test 4
7d left
بدء العمل
EM
employee1
قيد التنفيذ
1
test 2
8d left
وضع في قائمة الانتظار
تم الانتهاء
EM
employee1
مكتمل
1
test
8d left
بانتظار المراجعة
EM
employee1


تفاصيل المهمة
العنوان
test 2
الوصف
desc
الموظف
employee1
موعد التسليم
PM 2:01:00 ,6/28/2026
التعليقات
owner
comment from owner
o
employee1
comment from emp
e
اكتب تعليق...
ارسال
اغلاق

المهام التي قمت بتعينها

test DONE	28/06/2026, 14:00:00
test 2 IN_PROGRESS	28/06/2026, 14:01:00
Test 3 PENDING	28/06/2026, 00:25:00
Test 4 PENDING	28/06/2026, 00:25:00


e e
employee1
1
New task
You have been assigned a new task: test 5

PROBLEM 02

Tracking Employees' Work

1

Hard to Track

Many branches make it nearly impossible to track all work without a centralized dashboard.

2

Departments Multiply It

Each branch has multiple departments, multiplying the tracking challenge further.

3

Tracking Each User

Managers need to know each user's status: working, on holiday, or no longer active — and what they're working on now.

SOLUTION

An Interactive, Centralized Dashboard

CLARA gives managers one clean dashboard for the whole company — active users, partner companies, recent tasks, and recent documents at a glance.

248	32	164	57
Active Users	Active Companies	Tasks This Week	Docs Created

[Screenshot Placeholder]

Company-wide Dashboard

[Screenshot Placeholder]

Recent Tasks & Contracts

[Screenshot Placeholder]

Employee Status & Profiles

لوحة التحكم /

لوحة تحكم

الصف أ — الحجم والنطاق

إجمالي المستخدمين
11

الأقسام
4

الشركات النشطة
3

المهام (الكل)
7

الصف ب — النشاط والالتزام

مستخدمون نشطون
6

الفروع
2

عقود بانتظار الموافقة
0

مهام مفتوحة
5

«مستخدمون بلا قسم» (حاليًا 5) يعتمد على وجود `departmentId` في استجابة المستخدمين. مهام متأخرة: 0 — أدوار النظام: 4 — عقود نشطة (غير محذوفة): 1.

آخر المهام

عرض الكل ←

العنوان	الحالة	المعيّن إليه	الموعد
test 5	قيد الانتظار	employee1	٢٠٢٦/٦/٢٨، ٦:٥٩ م
Test 4	قيد الانتظار	employee1	٢٠٢٦/٦/٢٨، ١٢:٢٥ ص
Test 3	قيد الانتظار	employee1	٢٠٢٦/٦/٢٨، ١٢:٢٥ ص
test 4	مكتملة	employee3	٢٠٢٦/٦/٢٨، ٢:٠٢ م
test 3	قيد الانتظار	employee2	٢٠٢٦/٦/٢٨، ٢:٠٢ م
test 2	قيد التنفيذ	employee1	٢٠٢٦/٦/٢٨، ٢:٠١ م

آخر العقود

عرض الكل ←

النوع	الحالة	الشركة	تاريخ الإنشاء
عقد تأسيس	مسودة	شركة النور للتجارة والاستيراد	٢٠٢٦/٦/٢١، ١٠:٠٣ ص

لوحة التحكم / الموارد البشرية

دليل الموظفين

+ إضافة موظف

إجمالي الموظفين	نشط	في إجازة	الأقسام
11	6	0	4

ابحث بالاسم أو البريد... | كل الأقسام | كل الحالات

الموظف	المسمى الوظيفي	القسم	تاريخ التعيين	الحالة	الإجراءات
Adham Ehab domziii99@gmail.com AD	—	—	21/06/2026	نشط	
Adham Ehab adham.eh11@gmail.com AD	—	—	21/06/2026	نشط	
owner owner@g.com OW	صاحب العمل	الموارد البشرية	30/05/2026	نشط	
headBranch headBranch@g.com HE	مدير فرع	الموارد البشرية	30/05/2026	نشط	

Adham Ehab

Inactive (PENDING_VERIFICATION)

AD

—

—

معلومات الاتصال

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الرقم القومي

12345951255345

رقم الهاتف

01000000000

معلومات العمل

المسمى الوظيفي

—

القسم والفرع

—

تاريخ التعيين

21 Jun 2026

المهام التي قمت بتعينها

لا توجد مهام مسجلة لهذا الموظف.

المهام المرتبطة

لا توجد مهام مسجلة لهذا الموظف.

العقود التي أنشأها

لا توجد عقود مسجلة.

PROBLEM 03

Documents & Contract Management

A Archiving & Organization

Legal contracts are the backbone of the business, but offices hold tons of hard copies that are very difficult to manage and retrieve when needed.

B Creating & Editing Contracts

New contracts are created frequently and can take hours each — including searching through past hard copies. Doing this by hand is a major time sink.

So why make humans do such repetitive, time-consuming work — in the age of AI?

AI FEATURES

Turning hours of manual contract work into seconds

Made with GAMMA

The Contract Amendment Challenge

Company Request → Paper Contract (20+ pages) → Manual Search → Manual Editing 30-60 min → Risk of Human Error

≈ 5 Contracts Processed Per Day

- Time Consuming
- Error Prone
- Limited Productivity

Need for Intelligent OCR & Automated Contract Amendment

CORE: Clause-Level OCR & Retrieval

Goal: turn scanned Arabic contracts into structured JSON (fields & articles) despite OCR noise and long, repetitive boilerplate.

1 Phase 1 — OCR

Chandra-OCR-2

- Deskew, contrast, blank-page removal
- Extract text in overlapping 2-page chunks
- Repetition guard stops OCR looping

2 Phase 2 — Structured Extraction

Qwen (4-bit)

- Converts text chunks to JSON
- "Exact copy" rule — no hallucination
- Gap: still inherits OCR typos

3 Phase 3 — Template Correction

110 Templates

- From 32 ground-truth contracts
- Matches articles by content, not numbering
- Smart merge: fixes small diffs, protects large ones

Final Assembly: merges overlapping chunks, drops partial duplicates, keeps complete ones. Status: all 3 phases built, 110 templates approved.

CAT: Contract Amendment Transformer

1

Model

- Qwen3-4B (4-bit Quantized)
- Strong Arabic understanding
- Efficient fine-tuning, low hardware needs

2

Dataset

- 420 training samples
- 45 test samples
- Each: prompt + before/after clause

3

Training

- Framework: Unsloth
- Technique: LoRA fine-tuning
- Faster, lower VRAM, cost-efficient

4

Objective

- Input: clause + amendment request
- Output: action title + updated clause
- Preserves legal structure & consistency



Model Selection & Benchmarking

Evaluated Families

- **Llama**
- **Gemma**
- **Qwen**
- **Other Open-Source**

Qwen Variants Tested

Qwen2.5: 0.5B • 1.5B • 3B • 7B

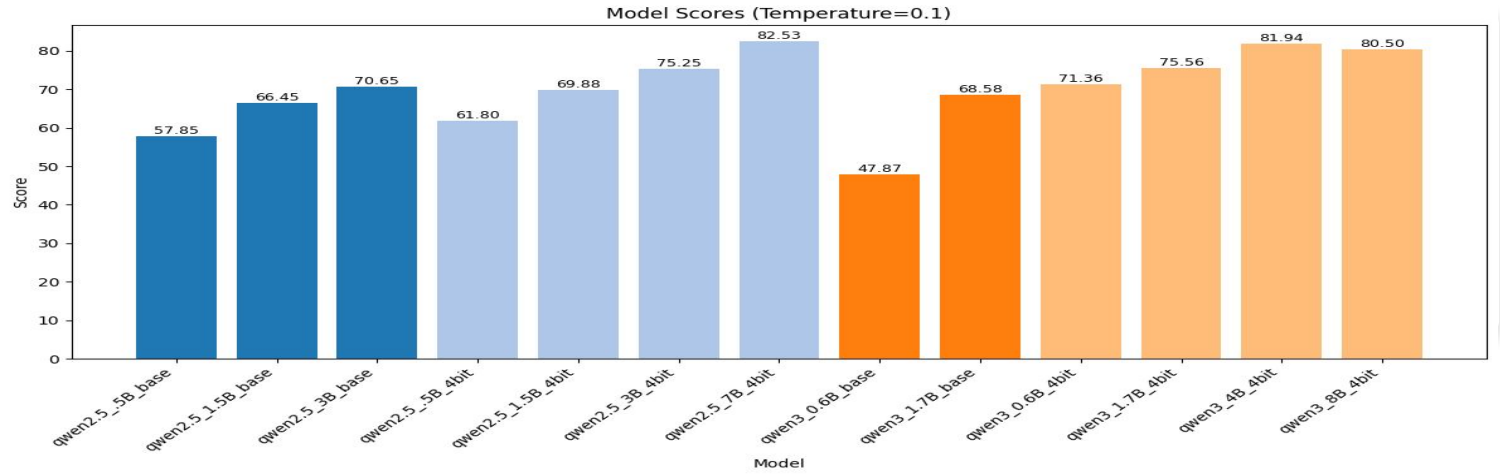
Qwen3: 0.6B • 1.7B • 4B • 8B

Selected: Qwen3-4B (4-bit)

Best overall score • stable editing • balanced accuracy vs resource use

Selection Criteria

- Amendment Accuracy
- Semantic Preservation
- Numerical Consistency
- Hardware Efficiency
- Fine-Tuning Performance



Limitation

GPU VRAM constrained larger experiments. With more VRAM, Qwen3-8B and beyond could be explored for potentially higher accuracy.

How the Scoring Pipeline Works

Comparing a fine-tuned model's prediction against a gold reference

ROUGE-L
Text Similarity

rouge-score (Google) — LCS-based overlap.

Semantic Similarity
Meaning Match

multilingual-e5-base embeddings, cosine similarity.

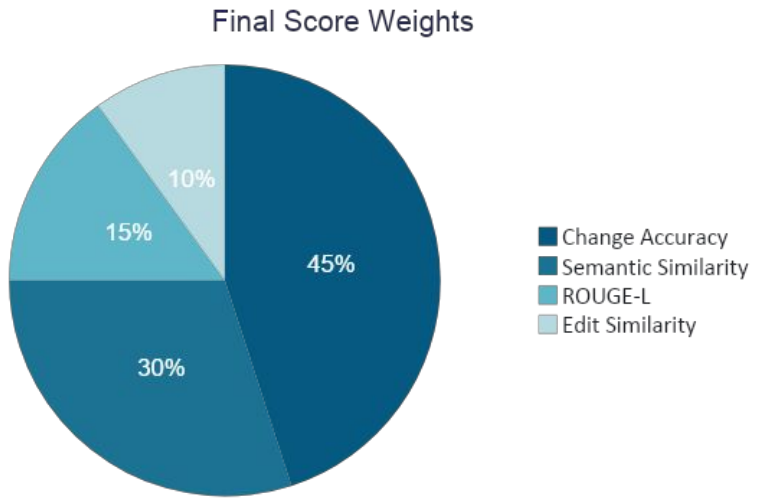
Edit Similarity
Character Match

difflib SequenceMatcher (Ratcliff/Obershelp).

Change Accuracy
Number Match

Regex + digit normalizer checks key numbers.

Four metrics are computed in parallel, then blended with weights into one 0-100 Final Score.



The Math Behind the Final Score

ROUGE-L

$$F = 2PR / (P + R)$$

P = LCS(pred,gold)/|pred| R = LCS(pred,gold)/|gold|

Semantic Similarity

$$sim = (e_pred \cdot e_gold) / (||e_pred|| \; ||e_gold||)$$

Cosine similarity x 100, using e5-base embeddings

Edit Similarity

$$ratio = 2M / T$$

M = matching chars, T = total chars (difflib)

Change Accuracy

$$Recall = |Nums_pred \cap Nums_gold| / |Nums_gold| \times 100$$

Checks whether key numbers were reproduced

$$Final = 0.45 \cdot CHA + 0.30 \cdot SS + 0.15 \cdot TS + 0.10 \cdot ES \quad \text{(gold has numbers)}$$

$$Final = 0.45 \cdot SS + 0.35 \cdot TS + 0.20 \cdot ES \quad \text{(no numbers)}$$

Clause-Level RAG Retrieval

Why Did We Use RAG?

Challenge

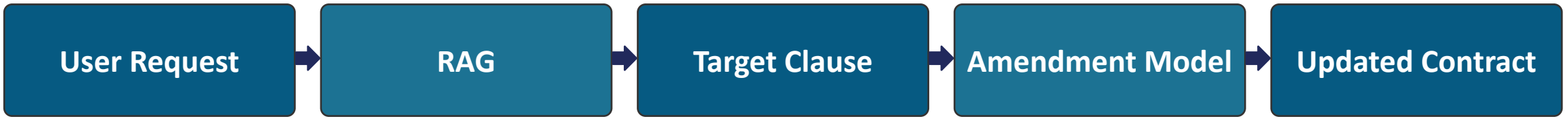
- Contracts may run 10-20+ pages
- Sending the full contract raises token cost
- Large context can hurt editing accuracy

Our Solution

- Split contract into individual clauses
- Retrieve only the clause tied to the request
- Send target clause to CAT, merge result back

Bonus: Clause Labeling

Each clause links to contract metadata (company name, duration, capital, address, business activities) — so CLARA can detect metadata-affecting edits and keep the JSON in sync.



Retrieval Models & Mathematical Foundation

Dense Retriever

intfloat/multilingual-e5-base

Converts Arabic clauses into dense semantic vectors; strong multilingual + Arabic retrieval accuracy. Stored & searched with FAISS.

Re-Ranker

BAAI/bge-reranker-v2-m3

Cross-encoder re-ranks top retrieved clauses by evaluating query and clause together, boosting precision over embedding search alone.

Mathematical Foundation

1. Embed clauses: $v_i = f(d_i)$
2. Normalize: $\hat{v}_i = v_i / ||v_i||, \quad \hat{q} = q / ||q||$
3. Similarity: $S(q,d_i) = \hat{q} \cdot \hat{v}_i$
4. Top-K Retrieval: $R = \text{TopK}(S(q,d_i))$
5. Re-Ranking: $r_i = g(q,d_i)$
6. Best Clause: $d^* = \text{argmax}(r_i)$

Why CAT & CORE Instead of General AI?

General AI (ChatGPT / Gemini)		CAT & CORE
✗ External cloud processing	**Privacy & Security**	✓ **Local office deployment**
✗ General purpose only	**Domain Knowledge**	✓ **Accounting & contract expertise**
✗ Entire contract context	**Document Understanding**	✓ **Clause-level retrieval (CORE)**
✗ May rewrite unrelated content	**Editing Precision**	✓ **Targeted, precise amendments**
✗ Pay per request / subscription	**Operational Cost**	✓ **One-time deployment cost**
✗ Output may vary	**Consistency**	✓ **Standardized amendment process**

Thank You

Questions & Discussion

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